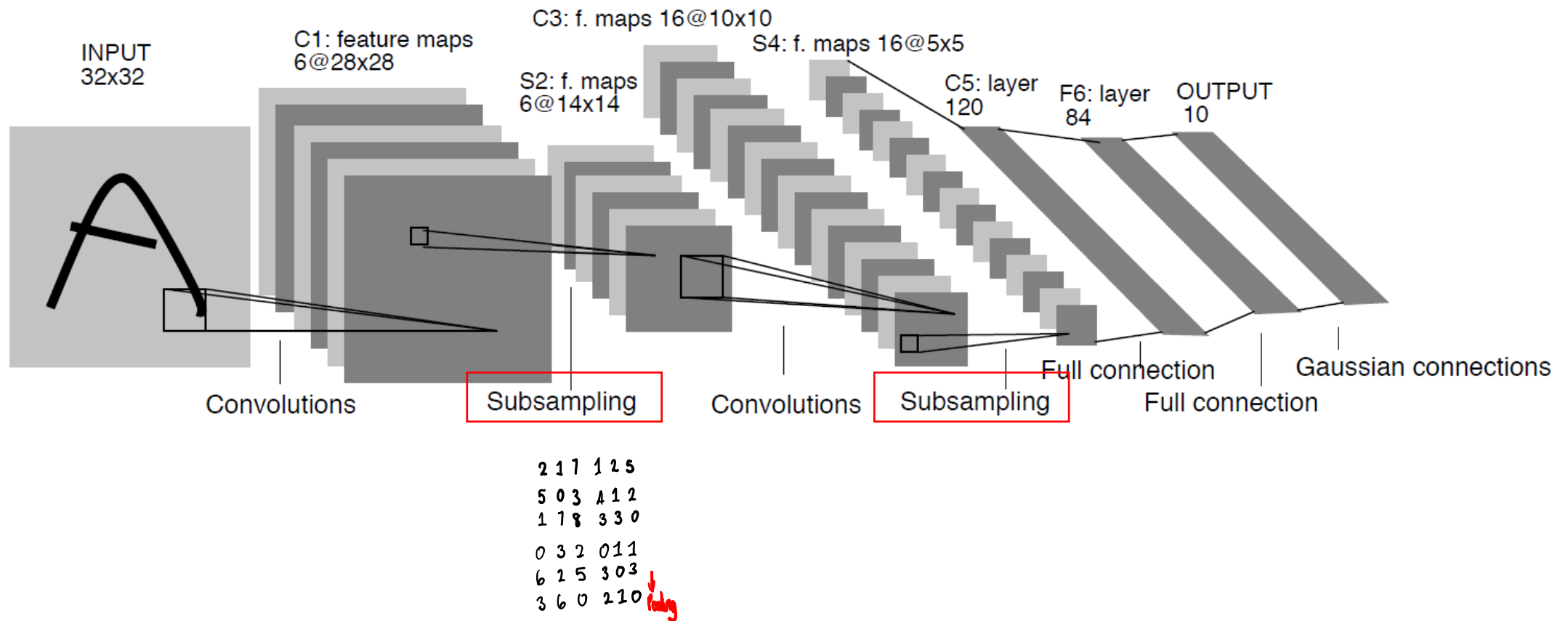


EEE4114F

Subsampling



Subsampling (Pooling) layers



Max-pooling

$$K = 3 \times 3, S = 3$$

2	1	7	1	2	5
5	0	3	4	1	2
1	7	8	3	3	0
0	3	2	0	1	1
6	2	5	3	0	3
3	6	0	2	1	0

output

8	5
6	3

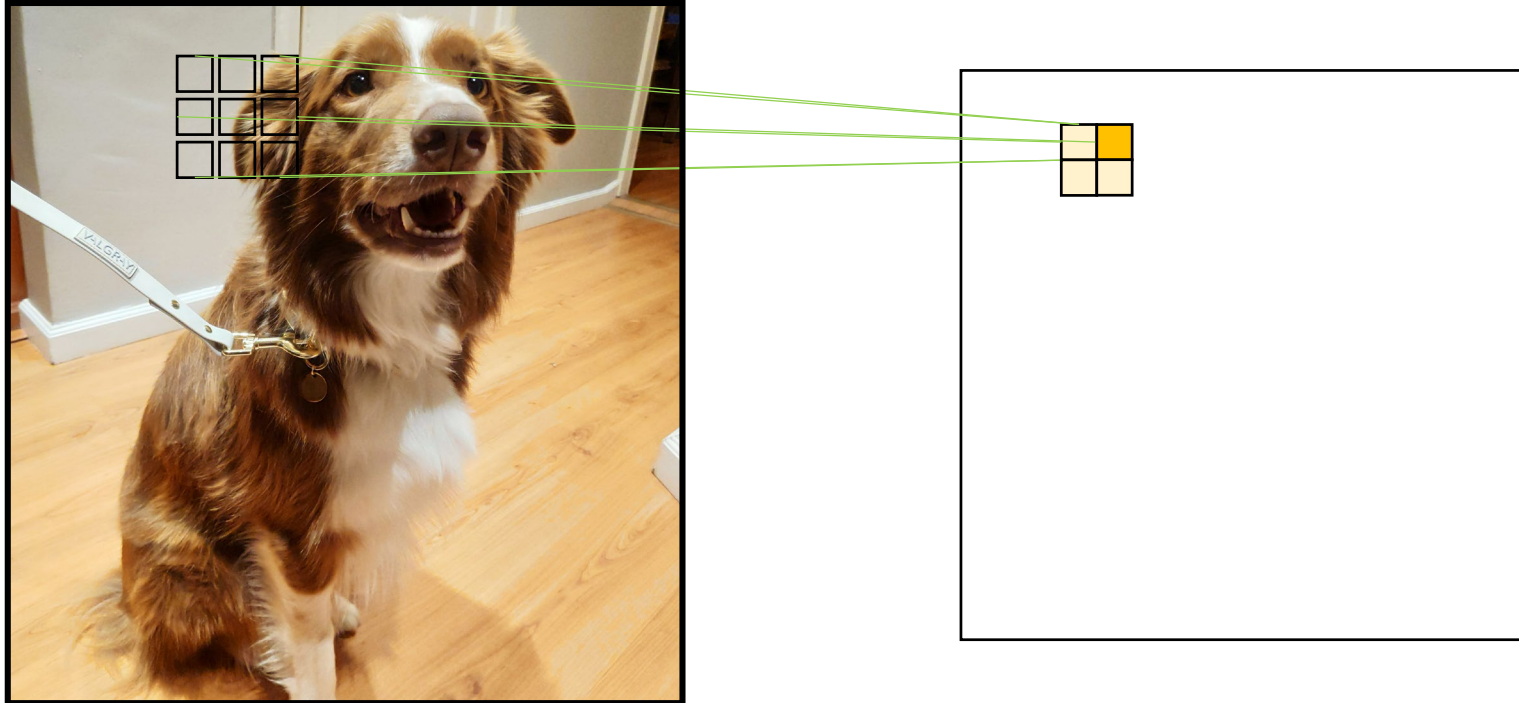
Size of input to MaxPool Padding Dilation Size of pooling kernel

$$O = \left\lfloor \frac{W_{in} + 2 * P - D \times (K - 1) - 1}{S} + 1 \right\rfloor$$

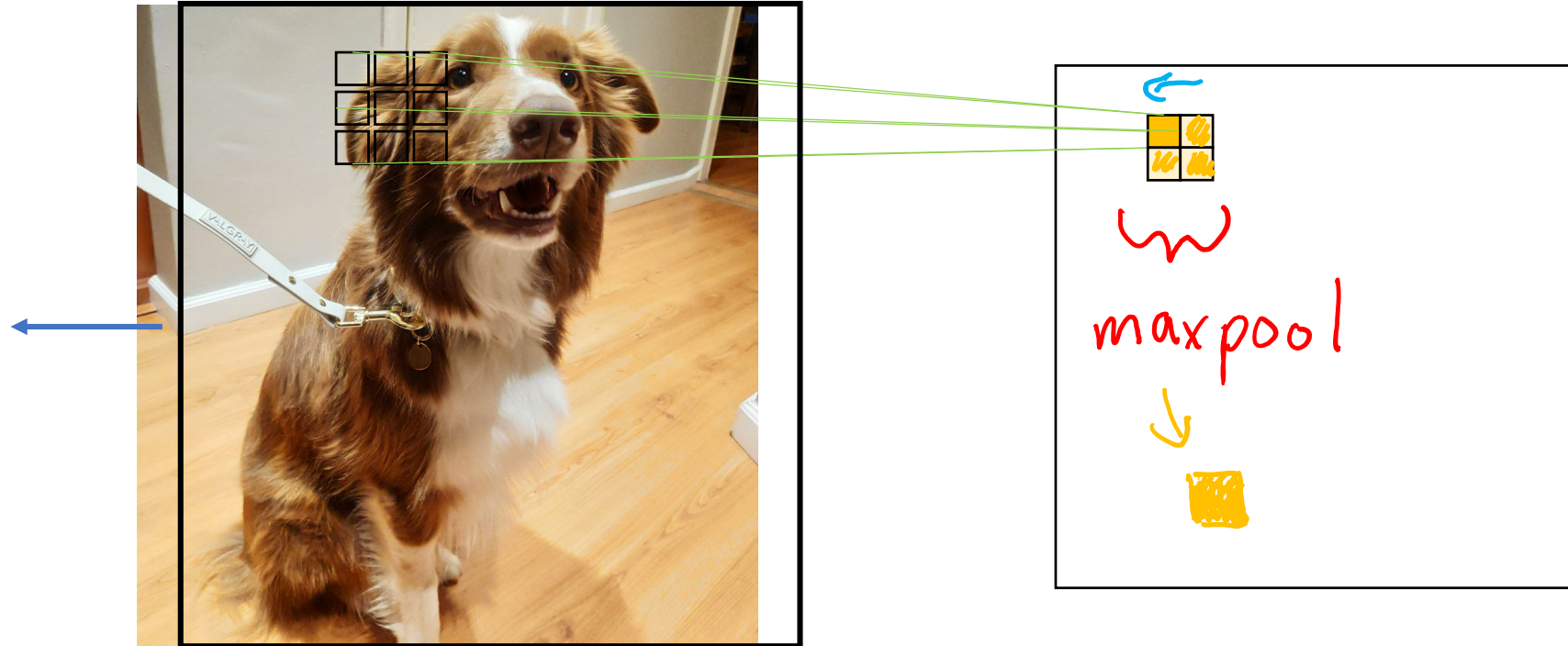
Stride of pooling kernel

$$\begin{aligned} O &= \left\lfloor \frac{6 - 1}{3} + 1 \right\rfloor \\ &= \left\lfloor 2.6667 \right\rfloor \\ &= 2 \end{aligned}$$

Invariance to translation



Invariance to translation



Next up:

CNN Example

